

In Exercises 23 and 24, mark each statement True or False. Justify each answer.

23a) A homogeneous equation is always consistent.

Solution: On Page 43, under the section Homogeneous Linear Systems, it is mentioned: "A system of linear equations is said to be homogeneous if it can be written in the form $Ax=0$, where A is an $m \times n$ matrix and 0 is the zero vector in $\mathbb{R}^m$. Such a system $Ax=0$ always has at least one solution, namely, $x=0$ (the zero vector in $\mathbb{R}^n$). From Page 4, we know that a system of linear equations is said to be consistent if it has at least one solution.

$\therefore$ The given statement is True $\square$

b) The equation $Ax=0$ gives an explicit description of its solution set.

Solution: On Page 45, just above the section "Parametric Vector Form" it is mentioned, just after Example 2 that the solution set of a homogeneous equation $Ax=0$ can always be expressed explicitly as $\text{Span}\{v_1, \dots, v_p\}$ for suitable vectors $v_1, \dots, v_p$.

$\therefore$ The given statement is True $\square$

c) The homogeneous equation $Ax=0$ has the trivial solution if and only if the equation has at least one free variable.

Solution: By Page 44, under the section "Homogeneous Linear Systems", it is mentioned that the homogeneous system $Ax=0$ always has the trivial solution $x=0$. The presence or absence of free variables does not matter.

$\therefore$ The given statement is False $\square$

d) The equation $x = p + tv$ describes a line through v parallel to p .

Solution: On page 46, the equation $x = p + tv$ (t in $\mathbb{R}$) is mentioned as eqn. 3. It is also mentioned that the eqn(3) is called the equation of the line through p parallel to v .

$\therefore$ The given statement is False $\square$